

AI Pricing in the Macroeconomy

Predictions for the Phillips Curve and Monetary Policy

CEBRA Annual Meeting, June 23 2026

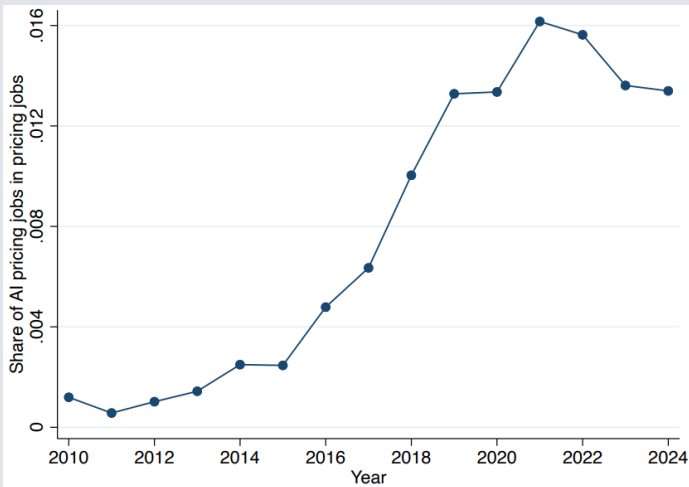
Jonathan J. Adams

Federal Reserve Bank of Kansas City

The views expressed herein are those of the author and do not necessarily represent the views of the Federal Reserve Bank of Kansas City or the Federal Reserve System.

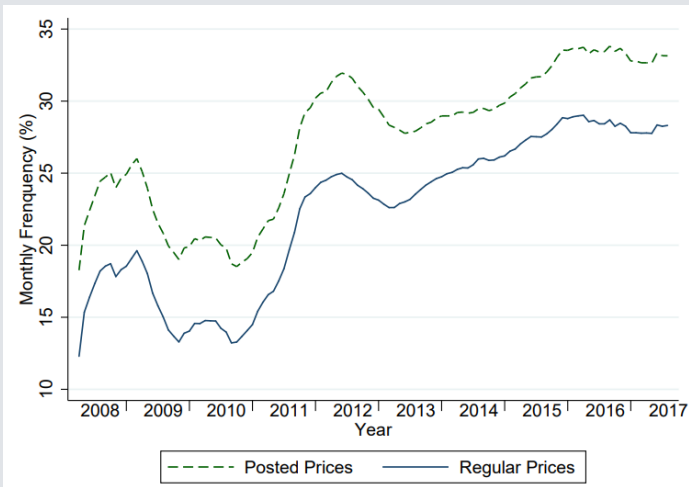
Motivation

Some Motivating Evidence



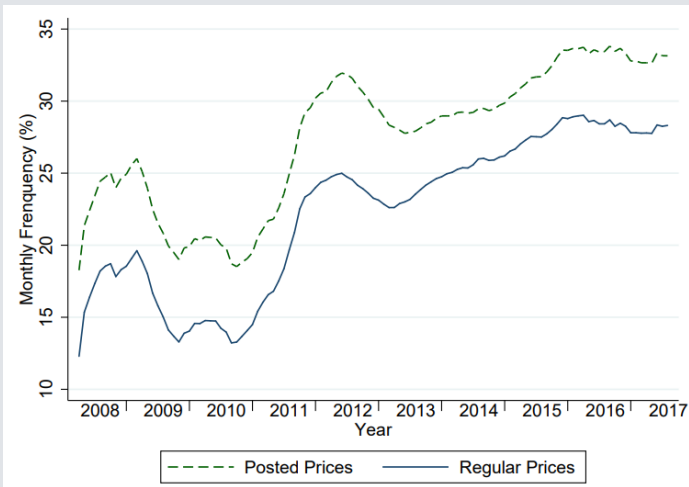
- AI (and algorithmic) pricing is becoming more common (Adams et al., 2026)

Some Motivating Evidence



- AI (and algorithmic) pricing is becoming more common (Adams et al., 2026)
- Frequency of price adjustment is rising (Cavallo, 2018)

Some Motivating Evidence



- AI (and algorithmic) pricing is becoming more common (Adams et al., 2026)
- Frequency of price adjustment is rising (Cavallo, 2018)
- Theory says the FPA matters for monetary policy

This Paper

- GE theory consistent with both trends and other stylized facts about AI pricing

This Paper

- GE theory consistent with both trends and other stylized facts about AI pricing
- A maximally tractable approach: similar to a menu cost model, so we can use existing results

This Paper

- GE theory consistent with both trends and other stylized facts about AI pricing
- A maximally tractable approach: similar to a menu cost model, so we can use existing results
- Characterize how AI pricing changes the Phillips curve and monetary policy

This Paper

- GE theory consistent with both trends and other stylized facts about AI pricing
- A maximally tractable approach: similar to a menu cost model, so we can use existing results
- Characterize how AI pricing changes the Phillips curve and monetary policy
- Provide 4 predictions for future AI pricing + monetary policy

Payoff: 4 Predictions

1. AI pricing will continue to rise as costs decline

Payoff: 4 Predictions

1. AI pricing will continue to rise as costs decline
2. Prices will become more flexible

Payoff: 4 Predictions

1. AI pricing will continue to rise as costs decline
2. Prices will become more flexible
3. The Phillips curve will become steeper

Payoff: 4 Predictions

1. AI pricing will continue to rise as costs decline
2. Prices will become more flexible
3. The Phillips curve will become steeper
4. Optimal monetary policy will respond more aggressively to inflation

What Is AI Pricing? (Conceptual)

- Firms set prices using a model of demand and costs

What Is AI Pricing? (Conceptual)

- Firms set prices using a model of demand and costs
- Model complexity is a choice

What Is AI Pricing? (Conceptual)

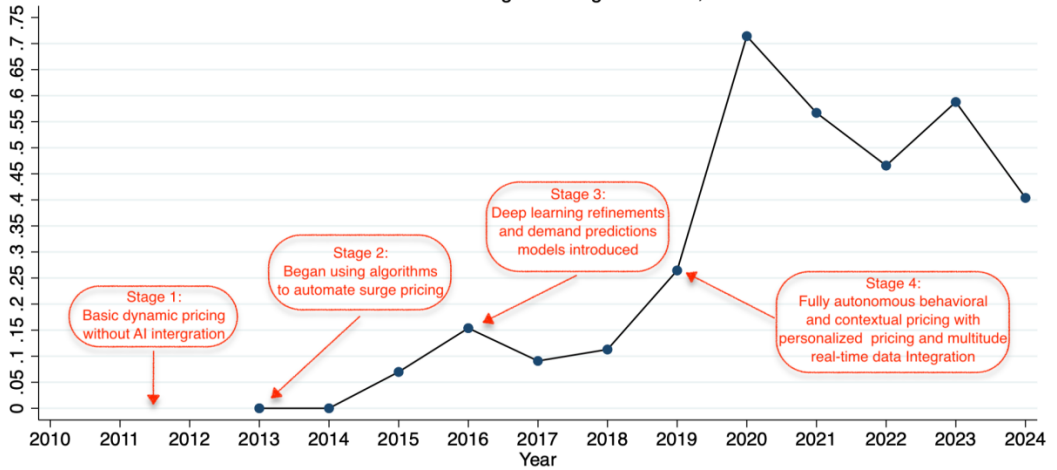
- Firms set prices using a model of demand and costs
- Model complexity is a choice
- Richer models are more costly

What Is AI Pricing? (Conceptual)

- Firms set prices using a model of demand and costs
- Model complexity is a choice
- Richer models are more costly
- But are more valuable for decision-making

What Is AI Pricing? (Example)

Uber: Fraction of AI Pricing in Pricing Job Posts, 2010-2024



A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)

A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)
- Composite facing Kimball-demand with i, j -specific demand+cost shocks.

Frictionless price:

$$\log P_{ijt}^* = (1 - \Omega) \log P_t + \Omega(\mu + \log W_t) + z_{ijt} \quad (1)$$

A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)
- Composite facing Kimball-demand with i, j -specific demand+cost shocks.

Frictionless price:

$$\log P_{ijt}^* = (1 - \Omega) \log P_t + \Omega(\mu + \log W_t) + z_{ijt} \quad (1)$$

- Composite shifter z_{ijt} is determined by many factors $k \in [0, 1]$:

$$z_{ijt} = \int_0^1 \hat{\beta}_{it}(k) F_{ij}(k) dk$$

A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)
- Composite facing Kimball-demand with i, j -specific demand+cost shocks.

Frictionless price:

$$\log P_{ijt}^* = (1 - \Omega) \log P_t + \Omega(\mu + \log W_t) + z_{ijt} \quad (1)$$

- Composite shifter z_{ijt} is determined by many factors $k \in [0, 1]$:

$$z_{ijt} = \int_0^1 \hat{\beta}_{it}(k) F_{ij}(k) dk$$

- Product/customer characteristics are summarized by factors $F_{ij}(k)$

A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)
- Composite facing Kimball-demand with i, j -specific demand+cost shocks.

Frictionless price:

$$\log P_{ijt}^* = (1 - \Omega) \log P_t + \Omega(\mu + \log W_t) + z_{ijt} \quad (1)$$

- Composite shifter z_{ijt} is determined by many factors $k \in [0, 1]$:

$$z_{ijt} = \int_0^1 \hat{\beta}_{it}(k) F_{ij}(k) dk$$

- Product/customer characteristics are summarized by factors $F_{ij}(k)$
- Coefficients $\hat{\beta}_{it}(k)$ drift between Poisson resets at rate ζ

A Factor Structure for Supply/Demand

- Firm i sells in markets $j \in [0, M]$ (locations, customers, etc.)
- Composite facing Kimball-demand with i, j -specific demand+cost shocks.

Frictionless price:

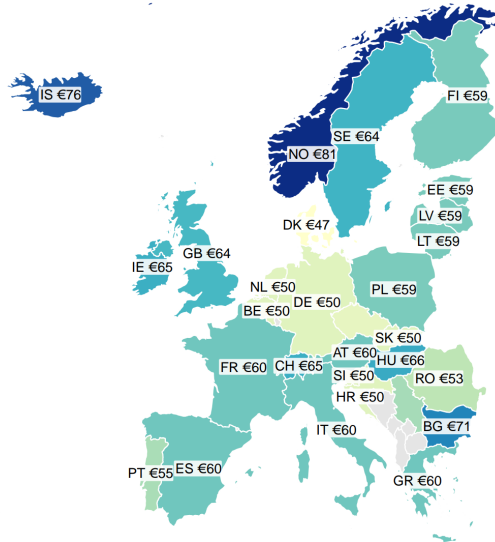
$$\log P_{ijt}^* = (1 - \Omega) \log P_t + \Omega(\mu + \log W_t) + z_{ijt} \quad (1)$$

- Composite shifter z_{ijt} is determined by many factors $k \in [0, 1]$:

$$z_{ijt} = \int_0^1 \hat{\beta}_{it}(k) F_{ij}(k) dk$$

- Product/customer characteristics are summarized by factors $F_{ij}(k)$
- Coefficients $\hat{\beta}_{it}(k)$ drift between Poisson resets at rate ζ
- Firm's model complexity = how many factors $F_{ij}(k)$ to use

Vad är AI-prissättning? (Exempel)



BILLY Bookcase
Price (€)



Source: IKEA

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

- Firms pay a **compute cost** to change $p_{it}(k)$

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

- Firms pay a **compute cost** to change $p_{it}(k)$
- Fundamentally different than a menu cost:

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

- Firms pay a **compute cost** to change $p_{it}(k)$
- Fundamentally different than a menu cost:
 - Invariant to the number of markets M

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

- Firms pay a **compute cost** to change $p_{it}(k)$
- Fundamentally different than a menu cost:
 - Invariant to the number of markets M
 - Scales with model complexity A_i

Price Setting with AI

- Firm i chooses model complexity $A_i \in [0, 1]$
- Its pricing rule uses coefficients $p_{it}(k)$ only for modeled factors:

$$\log P_{ijt} = \bar{P} + \int_0^{A_i} p_{it}(k) F_{ij}(k) dk$$

- Firms pay a **compute cost** to change $p_{it}(k)$
- Fundamentally different than a menu cost:
 - Invariant to the number of markets M
 - Scales with model complexity A_i
 - $\implies$ different behavior

Gap Representation

- *Gross gap* is factor k -specific:

$$g_{it}(k) = \underbrace{p_{it}(k) - \hat{\beta}_{it}(k)}_{x_{it}(k)} - \overbrace{\omega(k)a_t}^{\text{aggregates}}$$

Gap Representation

- *Gross gap* is factor k -specific:

$$g_{it}(k) = \underbrace{p_{it}(k) - \hat{\beta}_{it}(k)}_{x_{it}(k)} - \overbrace{\omega(k)a_t}^{\text{aggregates}}$$

- Firm i 's state is the *loading gap* $x_{it}(k)$: follows a Brownian motion between resets

Gap Representation

- *Gross gap* is factor k -specific:

$$g_{it}(k) = \underbrace{p_{it}(k) - \hat{\beta}_{it}(k)}_{x_{it}(k)} - \overbrace{\omega(k)a_t}^{\text{aggregates}}$$

- Firm i 's state is the *loading gap* $x_{it}(k)$: follows a Brownian motion between resets
- Integrating over factors k and products j , the firm's instantaneous loss is

$$\mathcal{L}_{it} = - \underbrace{\mathbf{BM} \int_0^{A_i} g_{it}(k)^2 dk}_{\text{modeled factors}} - \overbrace{\mathbf{BM}(1 - A_i)V_{\hat{z}}}^{\text{unmodeled factors}}$$

Firm i 's Decisionmaking

- Conceit for tractability: firms know $g_{it}(k)$

Firm i 's Decisionmaking

- Conceit for tractability: firms know $g_{it}(k)$
- Outer loop decision: choose complexity A_i , paying an increasing cost

Firm i 's Decisionmaking

- Conceit for tractability: firms know $g_{it}(k)$
- Outer loop decision: choose complexity A_i , paying an increasing cost
- The inner loop decision:

Firm i 's Decisionmaking

- Conceit for tractability: firms know $g_{it}(k)$
- Outer loop decision: choose complexity A_i , paying an increasing cost
- The inner loop decision:
 - Given complexity A_i , firms follow (s,S) strategy, with dynamic boundaries/reset points

Firm i 's Decisionmaking

- Conceit for tractability: firms know $g_{it}(k)$
- Outer loop decision: choose complexity A_i , paying an increasing cost
- The inner loop decision:
 - Given complexity A_i , firms follow (s,S) strategy, with dynamic boundaries/reset points
 - Looks like a menu cost model, but for a single factor k applied to M markets

Micro Predictions for AI Adoption

- More AI adoption when:

Micro Predictions for AI Adoption

- More AI adoption when:

1. firms sell more products/markets

Micro Predictions for AI Adoption

- **More AI adoption when:**

1. firms sell more products/markets
2. compute costs or fixed costs fall

Micro Predictions for AI Adoption

- **More AI adoption when:**

1. firms sell more products/markets
2. compute costs or fixed costs fall
3. demand is more elastic

Micro Predictions for AI Adoption

- **More AI adoption when:**
 1. firms sell more products/markets
 2. compute costs or fixed costs fall
 3. demand is more elastic
- We see (1) and (2) in the data (Adams et al., 2026)

Micro Predictions for AI Adoption

- **More AI adoption when:**
 1. firms sell more products/markets
 2. compute costs or fixed costs fall
 3. demand is more elastic
- We see (1) and (2) in the data (Adams et al., 2026)
- Similar predictions for *frequency of price adjustment* (ongoing work)

An AI Phillips Curve?

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics
- Menu cost model can be tractably approximated (Adams, 2026):

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics
- Menu cost model can be tractably approximated (Adams, 2026):
 - Linearized the mean field game, dynamic decisions

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics
- Menu cost model can be tractably approximated (Adams, 2026):
 - Linearized the mean field game, dynamic decisions
 - Discretize the continuous time model

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics
- Menu cost model can be tractably approximated (Adams, 2026):
 - Linearized the mean field game, dynamic decisions
 - Discretize the continuous time model
 - The *Primary Eigenvalue Decomposition* reduces the infinite dimensional state

Aggregation

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics
- Unmodeled factors $\in (A_i, 1]$: traditional Calvo dynamics
- Menu cost model can be tractably approximated (Adams, 2026):
 - Linearized the mean field game, dynamic decisions
 - Discretize the continuous time model
 - The *Primary Eigenvalue Decomposition* reduces the infinite dimensional state
 - Special case with zero trend inflation: isomorphic to Calvo

The AI Phillips Curve

Theorem: the *Primary Eigenvalue Decomposition* implies

$$\pi_t = \Lambda(A_i) mc_t + \beta \mathbb{E}_t \pi_{t+1}$$

with slope

$$\Lambda(A_i) \equiv \Omega \left(A_i \frac{(1 - \theta_{A_i})(1 - \beta \theta_{A_i})}{\theta_{A_i}} + (1 - A_i) \frac{(1 - \theta_{\neg A_i})(1 - \beta \theta_{\neg A_i})}{\theta_{\neg A_i}} \right)$$

$$\theta_{A_i} \equiv \exp \left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2} \right) \quad \theta_{\neg A_i} \equiv \exp(-\zeta)$$

► PED details

The AI Phillips Curve

Theorem: the *Primary Eigenvalue Decomposition* implies

$$\pi_t = \Lambda(A_i) mc_t + \beta \mathbb{E}_t \pi_{t+1}$$

with slope

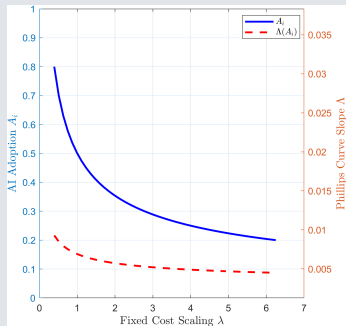
$$\Lambda(A_i) \equiv \Omega \left(A_i \frac{(1 - \theta_{A_i})(1 - \beta\theta_{A_i})}{\theta_{A_i}} + (1 - A_i) \frac{(1 - \theta_{\neg A_i})(1 - \beta\theta_{\neg A_i})}{\theta_{\neg A_i}} \right)$$

$$\theta_{A_i} \equiv \exp \left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2} \right) \quad \theta_{\neg A_i} \equiv \exp(-\zeta)$$

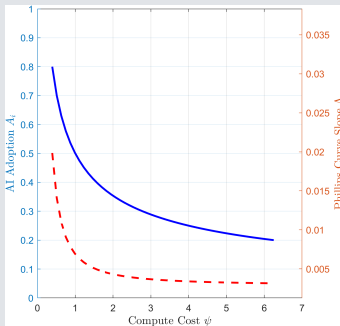
Slope $\Lambda(A_i)$ is increasing in A_i !

► PED details

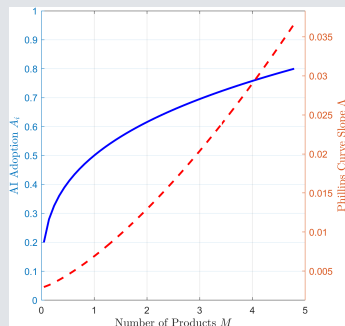
How Do AI Determinants Change the Slope?



(a) Fixed cost



(b) Compute cost



(c) Firm size

AI-NK Model

An AI-Pricing New Keynesian Model

- Replace the pricing block in a textbook New Keynesian model:

AI Phillips Curve: $\pi_t = \Lambda(A_i)\omega y_t + \beta \mathbb{E}_t \pi_{t+1} + u_t^c$

Euler equation: $\gamma y_t = \mathbb{E}_t[\gamma y_{t+1} - (i_t - \mathbb{E}_t \pi_{t+1})] + u_t^d$

Taylor rule: $i_t = \phi_\pi \pi_t + \phi_y y_t + u_t^m$

An AI-Pricing New Keynesian Model

- Replace the pricing block in a textbook New Keynesian model:

AI Phillips Curve: $\pi_t = \Lambda(A_i)\omega y_t + \beta \mathbb{E}_t \pi_{t+1} + u_t^c$

Euler equation: $\gamma y_t = \mathbb{E}_t[\gamma y_{t+1} - (i_t - \mathbb{E}_t \pi_{t+1})] + u_t^d$

Taylor rule: $i_t = \phi_\pi \pi_t + \phi_y y_t + u_t^m$

- Explore how A_i rising affects the macroeconomy

Calibration

Object	Value	Source
Discount rate ρ	0.04	Standard annual rate
Elasticity of substitution η	6	Standard
Strategic complementarity Ω	0.55	Gagliardone et al. (2025)
Profit loss curvature $\mathbf{B}$	33.3	Implied by η, Ω
Taylor Rule ϕ_π, ϕ_y	1.5, 0.5	Standard
Baseline AI adoption $\bar{A}$	0.5	Normalization
Diffusion variance σ_β^2	0.0073	Match moments
Poisson reset rate ζ	0.85	Match moments
Inaction boundary $\bar{x}$	0.088	Match moments
FPA F	1.26	French CPI microdata (Alvarez et al., 2024)
Std. dev. of price changes	0.076	French CPI microdata
Kurtosis of price changes	5.05	French CPI microdata

What Changes with AI Adoption?

- Higher AI adoption makes inflation more elastic to real activity

What Changes with AI Adoption?

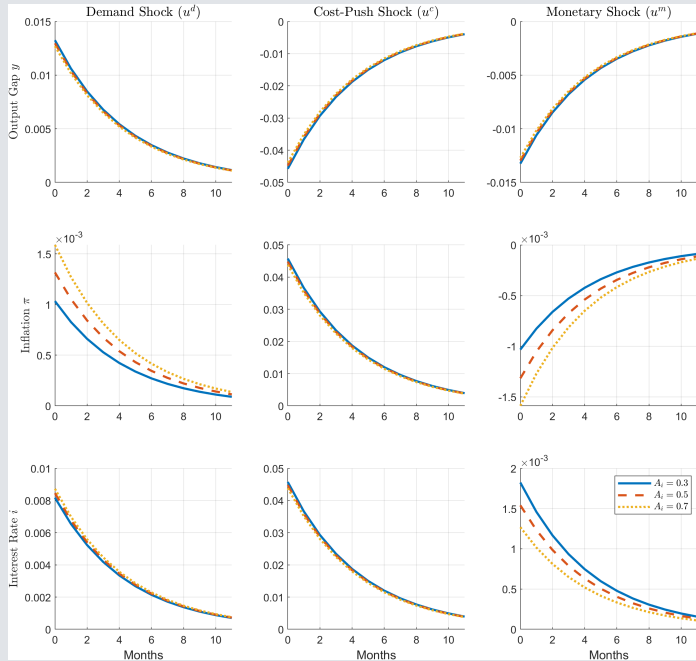
- Higher AI adoption makes inflation more elastic to real activity
- $\implies$ amplification of demand, monetary shocks (and other shocks in larger models)

What Changes with AI Adoption?

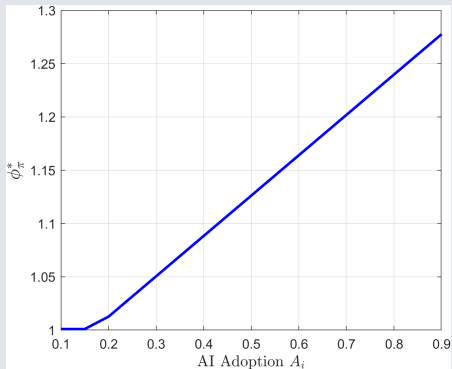
- Higher AI adoption makes inflation more elastic to real activity
- $\implies$ amplification of demand, monetary shocks (and other shocks in larger models)
- little change to cost-push shocks (enter the Phillips curve directly)

What Changes with AI Adoption?

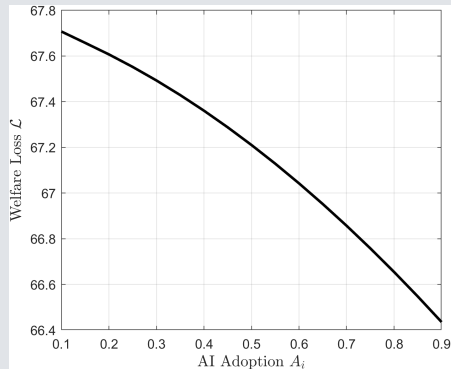
- Higher AI adoption makes inflation more elastic to real activity
- $\implies$ amplification of demand, monetary shocks (and other shocks in larger models)
- little change to cost-push shocks (enter the Phillips curve directly)
- Standard: changing the Phillips curve slope changes optimal monetary policy



Optimal Monetary Policy



(a) Optimal inflation response ϕ_π^*



(b) Minimized welfare loss $\mathcal{W}$

More AI pricing $\implies$ **more aggressive** monetary policy!

What's Next?

- The predictions so far:
 1. AI pricing will continue to rise as costs decline
 2. Prices will become more flexible
 3. The Phillips curve will become steeper
 4. Optimal monetary policy will respond more aggressively to inflation
- Gathering evidence relating AI use + pricing statistics
- New pricing theories!
- Really need better discipline on the AI pricing cost function (help!)

References

- Adams, Jonathan J.**, “Sticky Prices for Inflationary Economies: A Tractable Linear Approximation to Menu Cost Models with Trend Inflation,” *Federal Reserve Bank of Kansas City Working Paper*, March 2026, 26 (3).
- , **Min Fang, Zheng Liu, and Yajie Wang**, “The rise of AI pricing: Trends, driving forces, and implications for firm performance,” *Journal of Monetary Economics*, January 2026, 157, 103875.
- Alvarez, Fernando, Andrea Ferrara, Erwan Gautier, Hervé Le Bihan, and Francesco Lippi**, “Empirical Investigation of a Sufficient Statistic for Monetary Shocks,” *The Review of Economic Studies*, August 2024, p. rdae082.
- Cavallo, Alberto**, “More Amazon effects: online competition and pricing

PED Mapping

- For each modeled factor, zero-trend-inflation dynamics are approximated by:

$$x_{k,t}^* = (1 - \beta\theta_k)a_t + \beta\theta_k\mathbb{E}_t x_{k,t+1}^*$$

$$x_{k,t} = (1 - \theta_k)x_{k,t}^* + \theta_k x_{k,t-1}$$

- Modeled and unmodeled factors differ only in effective stickiness:

$$\theta_{A_i} = \exp\left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2}\right), \quad \theta_{\neg A_i} = \exp(-\zeta)$$

Price Change Moments

- Economy-wide price changes are a mixture:

$$F = A_i F^{CP} + (1 - A_i) F^C$$

$$\sigma_{\Delta p}^2 = \frac{A_i F^{CP} (\sigma_{\Delta p}^{CP})^2 + (1 - A_i) F^C (\sigma_{\Delta p}^C)^2}{F}$$

- Targets: frequency, standard deviation, kurtosis
- Calvo reset sizes have kurtosis 6; Calvo-plus lowers tail mass through bounded adjustments
- Baseline pins down $(\sigma_{\beta}^2, \zeta, \bar{x})$